

# Baseline Leaf Gate — Evaluation Results

*TinyLeafGate · 96×96 RGB · INT8 TFLite · classes: leaf / not\_leaf*

This report summarises the **deployed INT8 TFLite** gate that decides whether a camera frame is a leaf before the PlantVillage disease CNN runs. All figures below come from files already in this model folder: the Colab run-2 notes in *Colab Training Run History/TRAINING\_RESULTS.md*, the side-by-side error counts in the alt model's *ALT\_TRAINING\_REPORT.md*, and the recorded test split sizes of the leaf/not-leaf dataset. The confusion matrix is reconstructed from those counts (true-leaf rejects and false-leaf accepts) so that it matches the published per-class accuracies exactly.

## Sources used

Primary: *Colab Training Run History/TRAINING\_RESULTS.md* (run 2, 2026-07-28). Error counts *FN\_leaf / FP\_leaf*: *../Alt Leaf Gate Model/ALT\_TRAINING\_REPORT.md* §3.2 (same baseline TFLite on the same test split). Artifact: *aclis\_leaf\_gate\_96x\_full\_int8.tflite* (50.9 KB).

## 1. Overall accuracy

Overall accuracy is the share of test images whose predicted class matches the folder label. The number that matters for the STM32 cascade is the **INT8 TFLite** result on the held-out clean test set (n = 3,673). PyTorch FP32 is shown only as a quantisation-gap check: the two agree to 0.03 percentage points, well inside the notebook's 1% export gate.

|                            |                           |                     |              |
|----------------------------|---------------------------|---------------------|--------------|
| INT8 test accuracy         | PyTorch FP32              | Best validation     | Target       |
| <b>99.13%</b>              | <b>99.10%</b>             | <b>99.13%</b>       | <b>≥ 92%</b> |
| deployed model · n = 3,673 | same split · gap 0.03 pts | training checkpoint | PASS         |

| Metric                     | Value             | Notes                                           |
|----------------------------|-------------------|-------------------------------------------------|
| INT8 TFLite test accuracy  | 0.9913 (99.13%)   | Full test set via tf.lite.Interpreter           |
| PyTorch FP32 test accuracy | 0.9910 (99.10%)   | Same images, float student                      |
| INT8 – FP32  gap           | 0.0003 (0.03 pts) | Export abort threshold was 1%                   |
| Pred agreement INT8 ↔ PT   | 99.0–99.8%        | Same argmax as PyTorch                          |
| Best validation accuracy   | 0.9913            | End of 5 + 25 epoch schedule                    |
| TFLite size                | 50.9 KB           | Fits with 723 KB disease model in 1024 KB flash |

Table 1. Overall accuracy for the baseline TinyLeafGate (run 2).

## 2. Per-class accuracy

Per-class accuracy here is **recall**: of all images whose true label is that class, the fraction predicted correctly. The gate is slightly stronger on leaves than on non-leaves. That is the preferred bias for ACLIS: a missed leaf (false reject) drops a real plant from the disease path, while a false leaf only wastes a disease-CNN call.

| Class        | Test n | Correct | Errors | Accuracy | Meaning of an error               |
|--------------|--------|---------|--------|----------|-----------------------------------|
| leaf (0)     | 1,943  | 1,932   | 11     | 99.43%   | True leaf predicted not_leaf (FN) |
| not_leaf (1) | 1,730  | 1,709   | 21     | 98.79%   | Junk frame predicted leaf (FP)    |
| Overall      | 3,673  | 3,641   | 32     | 99.13%   | All test images                   |

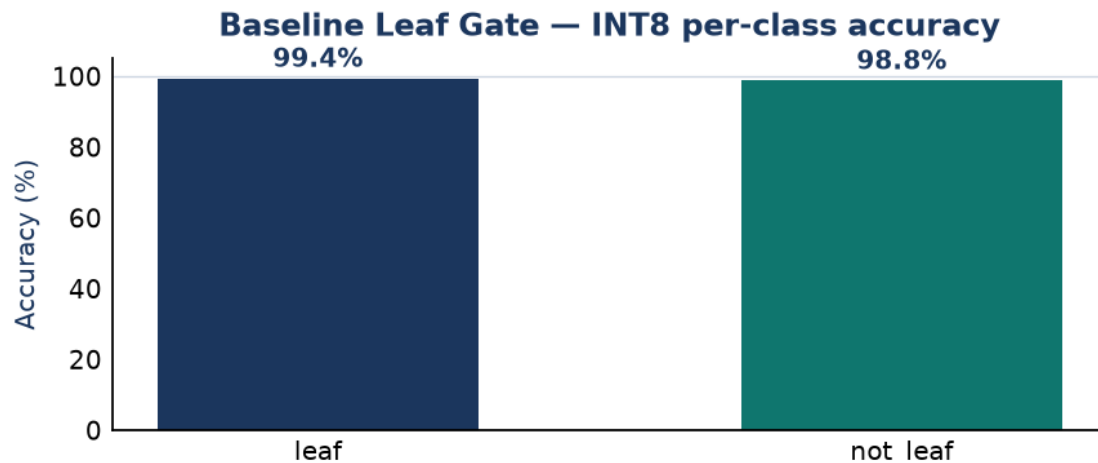


Figure 1. Clean INT8 per-class recall on the Kaggle leaf / not\_leaf test split.

### 3. Confusion matrix

Rows are the true label; columns are the predicted label. Diagonal cells are correct decisions. Off-diagonal cells are the only 32 mistakes on 3,673 images. Counts come from the recorded FN\_leaf = 11 and FP\_leaf = 21 together with the test-set sizes.

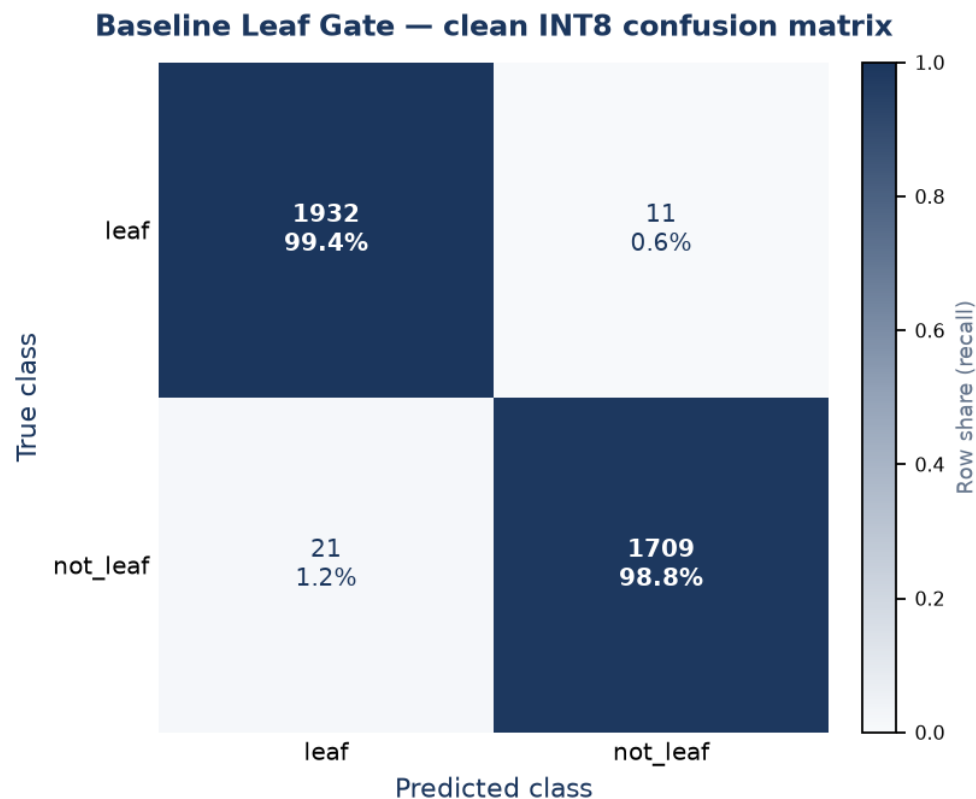


Figure 2. Confusion matrix, clean INT8 test. Cell text is count and row percentage.

| True \ Pred  | leaf  | not_leaf | Row total | Recall |
|--------------|-------|----------|-----------|--------|
| leaf         | 1,932 | 11       | 1,943     | 99.43% |
| not_leaf     | 21    | 1,709    | 1,730     | 98.79% |
| Column total | 1,953 | 1,720    | 3,673     | 99.13% |

Table 2. Numeric confusion matrix for the baseline INT8 model (clean test).

## How to read the matrix

**True leaf → predicted not\_leaf (11 images, 0.57% of leaves).** The gate rejects a real leaf, so the disease CNN never runs. This is the more costly error for the product.

**True not\_leaf → predicted leaf (21 images, 1.21% of non-leaves).** A non-plant frame is sent into the disease classifier. That costs ~800 ms on the MCU and may show a spurious disease label, but it does not drop a real leaf.

Net: the baseline gate is already near ceiling on this clean Kaggle split. Remaining errors are a handful of ambiguous frames, not a systematic class collapse.

## 4. Camera-stress proxy (same baseline TFLite)

The alt-recipe record also evaluated this baseline TFLite on a fixed camera-stress set (full-frame resize + blur + underexposure + JPEG q=55). That is not the primary Kaggle number, but it is the better proxy for OV2640 field frames. Under stress the baseline falls from 99.13% to **97.63%**, driven mostly by extra false leaves (FP\_leaf 21 → 70).

| Split              | Accuracy | leaf  | not_leaf | FN_leaf | FP_leaf |
|--------------------|----------|-------|----------|---------|---------|
| Clean INT8         | 99.13%   | 99.4% | 98.8%    | 11      | 21      |
| Camera-stress INT8 | 97.63%   | 99.1% | 96.0%    | 17      | 70      |

Table 3. Baseline INT8 on clean vs camera-stress (from the alt training report).